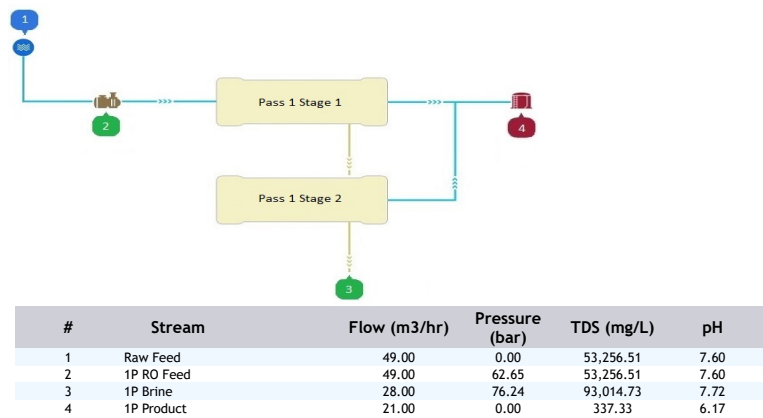


Solubility - Pass1

	Solubility calculation	
	Feed	Conc.
LSI	1.01	2
CaSO4	37.23 %	78.02 %
SrSO4	0 %	0 %
BaSO4	0 %	0 %
CaF2	0 %	0.01 %
SiO2	0 %	0 %
Stiff Davis Index	-0.6	0.5

Warnings - Pass1

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Project name: AFT_53PPM_Chile
Customer:
Username: Jiyoung Rowley

Water type: Antofagasta 53K
Flux loss per year: 7.00%
Salt passage increase: 7.00%

Membrane age: 0
Safety factor: 1

Overall System

Total permeate flow: 21 m3/hr
Raw water flow: 49 m3/hr
Total concentrate flow: 28 m3/hr
Overall recovery: 42.86 %

Water source: Seawater-Open Intake (SDI<5)
Raw water TDS: 53,250.15 mg/L
Feed osmotic pressure: 38.13 bar
Concentrate osmotic pressure: 66.59 bar

Feed pressure: 59.89 bar (1P)

System - Pass1

Permeate flow: 21 m3/hr
RO feed flow: 49 m3/hr
Concentrate flow: 28 m3/hr
Recovery: 42.86 %
Number of elements: 70
ERD type: None
Recirculation:

Average flux: 8.5 lmh
Water source: Seawater-Open Intake (SDI<5)
Feed TDS: 53,250.15 mg/L
Feed osmotic pressure: 38.13 bar
Concentrate osmotic pressure: 66.59 bar
Pump efficiency: 80 %

Temperature: 24 °C
Average NDP: 11.17 bar
Specific energy: 5.59 kWh/m³
Feed pressure: 59.89 bar
Permeate TDS: 352.6 mg/L
Fouling factor: 1

	# of vessels	# of elements	RO feed flow	Permeate flow	Conc. flow	RO feed pressure	Conc. pressure	Vessel DP	Boost pressure	Back pressure	Inter-stage pressure loss	Average flux	Perm. TDS
			m3/hr	m3/hr	m3/hr	bar	bar	bar	bar	bar	bar	lmh	mg/L
Stage 1	6	7	49	13.88	35.12	59.89	58.84	1.06	0	1	0	9.36	288.64
Stage 2	4	7	35.12	7.14	27.98	74.84	73.52	1.32	16	1	0	7.23	476.82

Water Analysis - Pass1

Species	Raw water	Adjusted feed	Conc.			Permeate		
			Stage1	Stage2	Composite	Stage1	Stage2	Composite
Ammonium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Sodium	16,579.05	16,579.05	23,088.29	28,940.92	28,940.92	103.69	171.39	126.70
Potassium	584.10	584.10	813.17	1,019.02	1,019.02	4.30	7.11	5.26
Magnesium	1,802.68	1,802.68	2,513.93	3,154.91	3,154.91	2.45	4.06	3.00
Calcium	623.39	623.39	869.35	1,091.02	1,091.02	0.84	1.39	1.02
Strontium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Barium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fluoride	0.01	0.01	0.01	0.02	0.02	0.00	0.00	0.00
Chloride	29,229.23	29,229.23	40,710.45	51,036.18	51,036.18	169.43	280.08	207.04
Sulfate	4,145.74	4,145.74	5,782.74	7,258.62	7,258.62	2.38	3.93	2.91
Nitrate	0.20	0.20	0.28	0.34	0.34	0.01	0.01	0.01
Carbonate	3.18	3.18	4.43	5.56	5.56	0.00	0.00	0.00
Bicarbonate	273.00	273.00	379.33	474.57	474.57	3.88	6.39	4.73
Boron	7.48	7.48	9.78	11.66	11.66	1.66	2.44	1.92
Bromide	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Silica	2.10	2.10	2.93	3.67	3.67	0.01	0.01	0.01
CO2	7.08	7.08	7.08	7.08	7.08	7.08	7.08	7.08
TDS	53,250.15	53,250.15	74,174.70	92,996.50	92,996.50	288.64	476.82	352.60
pH	7.60	7.60	7.73	7.82	7.82	5.93	6.14	6.00

Within Vessels - Pass1

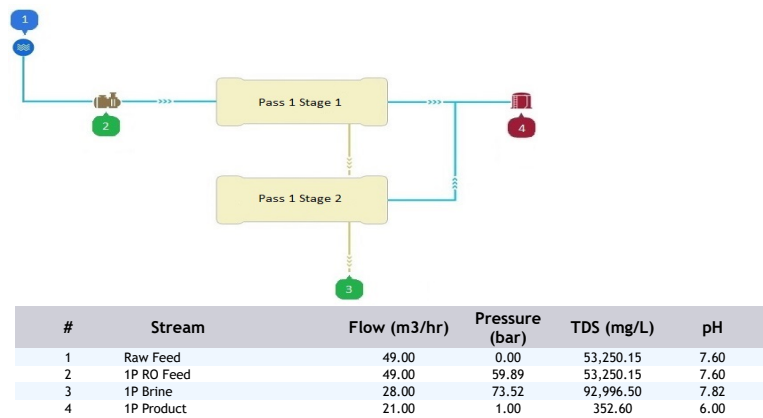
	Position	RO feed flow	Permeate flow	Flux	Element recovery	Element DP	Net driving pressure	Polarization	Feed TDS	Perm. TDS
		m3/hr	m3/hr	lmh	%	bar	bar		mg/L	mg/L
Stage 1										
LG SW 400 R G2 UHP	1	8.17	0.63	17.75	7.68	0.20	14.72	1.11	53,243.49	135.46
LG SW 400 R G2 UHP	2	7.54	0.49	13.77	6.45	0.17	12.09	1.09	57,658.57	184.40
LG SW 400 R G2 UHP	3	7.05	0.38	10.63	5.32	0.16	9.87	1.08	61,618.49	249.43
LG SW 400 R G2 UHP	4	6.68	0.29	8.20	4.33	0.14	8.03	1.06	65,065.71	334.47
LG SW 400 R G2 UHP	5	6.39	0.22	6.35	3.51	0.14	6.52	1.05	67,998.72	443.93
LG SW 400 R G2 UHP	6	6.16	0.17	4.94	2.83	0.13	5.30	1.04	70,453.73	582.17
LG SW 400 R G2 UHP	7	5.99	0.14	3.88	2.29	0.12	4.32	1.03	72,488.56	753.75
Stage 2										
LG SW 400 R G2 UHP	1	8.78	0.47	13.24	5.32	0.23	15.06	1.08	74,166.75	240.12
LG SW 400 R G2 UHP	2	8.31	0.36	10.32	4.38	0.21	12.76	1.06	78,324.07	319.11
LG SW 400 R G2 UHP	3	7.95	0.29	8.08	3.59	0.19	10.84	1.05	81,898.81	418.85
LG SW 400 R G2 UHP	4	7.66	0.23	6.39	2.94	0.18	9.23	1.04	84,933.36	542.29
LG SW 400 R G2 UHP	5	7.44	0.18	5.10	2.42	0.18	7.88	1.03	87,491.44	692.00
LG SW 400 R G2 UHP	6	7.26	0.15	4.11	2.00	0.17	6.76	1.03	89,642.99	869.96
LG SW 400 R G2 UHP	7	7.11	0.12	3.36	1.67	0.16	5.82	1.02	91,455.15	1,077.42

Solubility - Pass1

	Solubility calculation	
	Feed	Conc.
LSI	1.08	2.11
CaSO4	37.24 %	78.04 %
SrSO4	0 %	0 %
BaSO4	0 %	0 %
CaF2	0 %	0.01 %
SiO2	0 %	0 %
Stiff Davis Index	-0.55	0.59

Warnings - Pass1

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Project name: AFT_53PPM_Chile
Customer:
Username: Jiyoung Rowley

Water type: Antofagasta 53K
Flux loss per year: 7.00%
Salt passage increase: 7.00%

Membrane age: 1
Safety factor: 1

Overall System

Total permeate flow: 21 m3/hr
Raw water flow: 49 m3/hr
Total concentrate flow: 28 m3/hr
Overall recovery: 42.86 %

Water source: Seawater-Open Intake (SDI<5)
Raw water TDS: 53,250.15 mg/L
Feed osmotic pressure: 38.13 bar
Concentrate osmotic pressure: 66.58 bar

Feed pressure: 60.5 bar (1P)

System - Pass1

Permeate flow: 21 m3/hr
RO feed flow: 49 m3/hr
Concentrate flow: 28 m3/hr
Recovery: 42.86 %
Number of elements: 70
ERD type: None
Recirculation:

Average flux: 8.5 lmh
Water source: Seawater-Open Intake (SDI<5)
Feed TDS: 53,250.15 mg/L
Feed osmotic pressure: 38.13 bar
Concentrate osmotic pressure: 66.58 bar
Pump efficiency: 80 %

Temperature: 24 °C
Average NDP: 11.82 bar
Specific energy: 5.64 kWh/m³
Feed pressure: 60.5 bar
Permeate TDS: 376.7 mg/L
Fouling factor: 0.93

	# of vessels	# of elements	RO feed flow	Permeate flow	Conc. flow	RO feed pressure	Conc. pressure	Vessel DP	Boost pressure	Back pressure	Inter-stage pressure loss	Average flux	Perm. TDS
			m3/hr	m3/hr	m3/hr	bar	bar	bar	bar	bar	bar	lmh	mg/L
Stage 1	6	7	49	13.91	35.09	60.5	59.44	1.06	0	1	0	9.38	307.69
Stage 2	4	7	35.09	7.11	27.98	75.44	74.12	1.32	16	1	0	7.2	511.62

Water Analysis - Pass1

Species	Raw water	Adjusted feed	Conc.			Permeate		
			Stage1	Stage2	Composite	Stage1	Stage2	Composite
Ammonium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Sodium	16,579.05	16,579.05	23,105.76	28,933.99	28,933.99	110.54	183.92	135.37
Potassium	584.10	584.10	813.76	1,018.74	1,018.74	4.59	7.63	5.62
Magnesium	1,802.68	1,802.68	2,516.07	3,154.72	3,154.72	2.61	4.35	3.20
Calcium	623.39	623.39	870.09	1,090.95	1,090.95	0.89	1.49	1.09
Strontium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Barium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fluoride	0.01	0.01	0.01	0.02	0.02	0.00	0.00	0.00
Chloride	29,229.23	29,229.23	40,741.62	51,024.85	51,024.85	180.62	300.55	221.21
Sulfate	4,145.74	4,145.74	5,787.76	7,258.38	7,258.38	2.53	4.22	3.10
Nitrate	0.20	0.20	0.28	0.34	0.34	0.01	0.02	0.01
Carbonate	3.18	3.18	4.44	5.56	5.56	0.00	0.00	0.00
Bicarbonate	273.00	273.00	379.56	474.32	474.32	4.13	6.86	5.06
Boron	7.48	7.48	9.75	11.58	11.58	1.74	2.56	2.02
Bromide	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Silica	2.10	2.10	2.93	3.67	3.67	0.01	0.01	0.01
CO2	7.08	7.08	7.08	7.08	7.08	7.08	7.08	7.08
TDS	53,250.15	53,250.15	74,232.04	92,977.12	92,977.12	307.69	511.62	376.70
pH	7.60	7.60	7.73	7.82	7.82	5.96	6.17	6.03

Within Vessels - Pass1

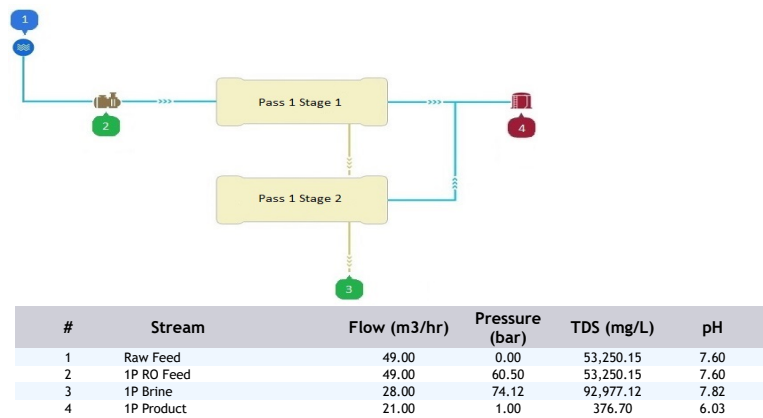
	Position	RO feed flow	Permeate flow	Flux	Element recovery	Element DP	Net driving pressure	Polarization	Feed TDS	Perm. TDS
		m3/hr	m3/hr	lmh	%	bar	bar		mg/L	mg/L
Stage 1										
LG SW 400 R G2 UHP	1	8.17	0.61	17.37	7.51	0.20	15.48	1.11	53,243.49	147.56
LG SW 400 R G2 UHP	2	7.55	0.48	13.64	6.37	0.17	12.85	1.09	57,553.16	198.36
LG SW 400 R G2 UHP	3	7.07	0.38	10.66	5.32	0.16	10.61	1.08	61,457.63	265.21
LG SW 400 R G2 UHP	4	6.70	0.29	8.32	4.39	0.14	8.73	1.06	64,895.79	351.87
LG SW 400 R G2 UHP	5	6.40	0.23	6.51	3.59	0.14	7.17	1.05	67,856.78	462.44
LG SW 400 R G2 UHP	6	6.17	0.18	5.12	2.93	0.13	5.89	1.04	70,365.99	601.48
LG SW 400 R G2 UHP	7	5.99	0.14	4.06	2.39	0.12	4.84	1.03	72,468.97	771.99
Stage 2										
LG SW 400 R G2 UHP	1	8.77	0.46	12.90	5.19	0.22	15.79	1.08	74,224.17	264.67
LG SW 400 R G2 UHP	2	8.32	0.36	10.17	4.32	0.21	13.50	1.06	78,273.45	347.67
LG SW 400 R G2 UHP	3	7.96	0.28	8.05	3.57	0.19	11.57	1.05	81,788.00	451.59
LG SW 400 R G2 UHP	4	7.67	0.23	6.42	2.95	0.18	9.94	1.04	84,800.70	579.20
LG SW 400 R G2 UHP	5	7.45	0.18	5.17	2.45	0.18	8.56	1.03	87,364.99	732.92
LG SW 400 R G2 UHP	6	7.26	0.15	4.21	2.04	0.17	7.41	1.03	89,541.85	914.55
LG SW 400 R G2 UHP	7	7.12	0.12	3.46	1.72	0.16	6.44	1.02	91,391.40	1,125.00

Solubility - Pass1

	Solubility calculation	
	Feed	Conc.
LSI	1.08	2.11
CaSO4	37.24 %	78.04 %
SrSO4	0 %	0 %
BaSO4	0 %	0 %
CaF2	0 %	0.01 %
SiO2	0 %	0 %
Stiff Davis Index	-0.55	0.59

Warnings - Pass1

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Project name:
Customer:
Username:

Water type: Antofagasta 53k
Flux loss per year: 7.00%
Salt passage increase: 7.00%

Membrane age: 5
Safety factor: 1

Overall System

Total permeate flow: 21 m3/hr
Raw water flow: 49 m3/hr
Total concentrate flow: 28 m3/hr
Overall recovery: 42.86 %

Water source: Seawater-Open Intake (SDI<5)
Raw water TDS: 53,257.46 mg/L
Feed osmotic pressure: 38.13 bar
Concentrate osmotic pressure: 66.52 bar

Feed pressure: 61.33 bar (1P)

Specific energy: 5.89 kwh/m3

System - Pass1

Permeate flow: 21 m3/hr
RO feed flow: 49 m3/hr
Concentrate flow: 28 m3/hr
Recovery: 42.86 %
Number of elements: 70
ERD type: None
Recirculation:

Average flux: 8.07 lmh
Water source: Seawater-Open Intake (SDI<5)
Feed TDS: 53,257.46 mg/L
Feed osmotic pressure: 38.13 bar
Concentrate osmotic pressure: 66.52 bar
Pump efficiency: 80 %

Temperature: 24 °C
Average NDP: 14.01 bar
Specific energy: 5.89 kwh/m3
Feed pressure: 61.33 bar
Permeate TDS: 491.25 mg/L
Fouling factor: 0.7

	# of vessels	# of elements	RO feed flow	Permeate flow	Conc. flow	RO feed pressure	Conc. pressure	Vessel DP	Boost pressure	Back pressure	Inter-stage pressure loss	Average flux	Perm. TDS
			m3/hr	m3/hr	m3/hr	bar	bar	bar	bar	bar	bar	lmh	mg/L
Stage 1	6	7	49	13.97	35.02	61.33	60.26	1.07	0	0	0	8.95	399.22
Stage 2	4	7	35.02	7.05	27.98	76.26	74.94	1.32	16	0	0	6.77	673.68

Water Analysis - Pass1

Species	Raw water	Adjusted feed	Conc.			Permeate		
			Stage1	Stage2	Composite	Stage1	Stage2	Composite
Ammonium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Sodium	16,582.00	16,582.00	23,139.28	28,907.65	28,907.65	144.23	243.56	177.54
Potassium	584.20	584.20	814.86	1,017.61	1,017.61	5.98	10.11	7.37
Magnesium	1,803.00	1,803.00	2,520.89	3,154.55	3,154.55	3.41	5.76	4.20
Calcium	623.50	623.50	871.76	1,090.90	1,090.90	1.15	1.95	1.42
Strontium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Barium	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Fluoride	0.01	0.01	0.01	0.02	0.02	0.00	0.00	0.00
Chloride	29,223.99	29,223.99	40,788.11	50,964.34	50,964.34	235.05	396.95	289.34
Sulfate	4,145.00	4,145.00	5,797.19	7,256.34	7,256.34	3.30	5.58	4.06
Nitrate	0.20	0.20	0.28	0.34	0.34	0.01	0.02	0.01
Carbonate	12.98	12.98	18.15	22.72	22.72	0.01	0.02	0.01
Bicarbonate	273.00	273.00	380.33	474.47	474.47	3.95	6.66	4.86
Boron	7.48	7.48	9.62	11.27	11.27	2.11	3.06	2.43
Bromide	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Silica	2.10	2.10	2.94	3.67	3.67	0.01	0.01	0.01
CO2	3.89	3.89	3.89	3.89	3.89	3.89	3.89	3.89
TDS	53,257.46	53,257.46	74,343.41	92,903.88	92,903.88	399.22	673.68	491.25
pH	7.60	7.60	7.67	7.72	7.72	6.24	6.45	6.32

Within Vessels - Pass1

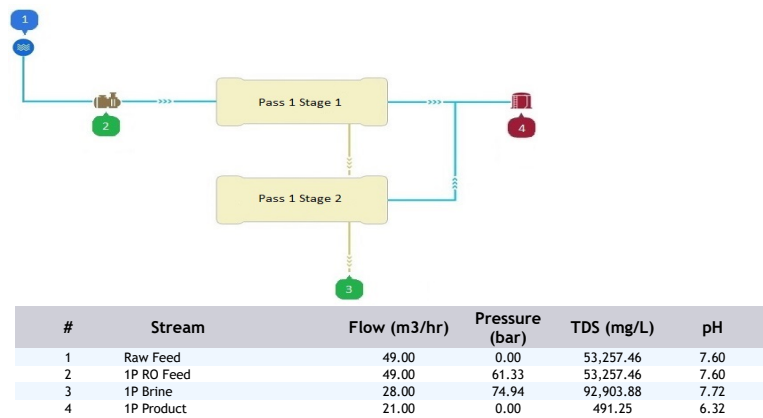
	Position	RO feed flow	Permeate flow	Flux	Element recovery	Element DP	Net driving pressure	Polarization	Feed TDS	Perm. TDS
		m3/hr	m3/hr	lmh	%	bar	bar		mg/L	mg/L
Stage 1										
LG SW 400 R G2 UHP	1	8.17	0.57	15.42	7.02	0.20	17.99	1.10	53,250.80	204.06
LG SW 400 R G2 UHP	2	7.59	0.47	12.55	6.14	0.18	15.38	1.08	57,253.27	264.63
LG SW 400 R G2 UHP	3	7.13	0.38	10.16	5.30	0.16	13.07	1.07	60,984.09	342.26
LG SW 400 R G2 UHP	4	6.75	0.30	8.20	4.52	0.15	11.06	1.06	64,377.68	440.53
LG SW 400 R G2 UHP	5	6.44	0.25	6.62	3.82	0.14	9.33	1.05	67,402.24	563.27
LG SW 400 R G2 UHP	6	6.20	0.20	5.36	3.21	0.13	7.87	1.04	70,055.19	714.36
LG SW 400 R G2 UHP	7	6.00	0.16	4.35	2.70	0.12	6.64	1.03	72,355.52	897.40
Stage 2										
LG SW 400 R G2 UHP	1	8.76	0.42	11.36	4.82	0.23	18.24	1.07	74,335.50	370.78
LG SW 400 R G2 UHP	2	8.33	0.34	9.26	4.13	0.21	15.97	1.06	78,081.35	470.89
LG SW 400 R G2 UHP	3	7.99	0.28	7.57	3.52	0.20	13.99	1.05	81,424.03	593.09
LG SW 400 R G2 UHP	4	7.71	0.23	6.22	3.00	0.19	12.29	1.04	84,373.67	739.92
LG SW 400 R G2 UHP	5	7.48	0.19	5.14	2.55	0.18	10.82	1.03	86,957.50	913.44
LG SW 400 R G2 UHP	6	7.29	0.16	4.28	2.18	0.17	9.55	1.03	89,212.30	1,115.07
LG SW 400 R G2 UHP	7	7.13	0.13	3.60	1.88	0.16	8.47	1.02	91,178.33	1,345.40

Solubility - Pass1

	Solubility calculation	
	Feed	Conc.
LSI	1.09	2.08
CaSO4	37.23 %	78.05 %
SrSO4	0 %	0 %
BaSO4	0 %	0 %
CaF2	0 %	0.01 %
SiO2	0 %	0 %
Stiff Davis Index	-0.53	0.56

Warnings - Pass1

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Technical Proposal

8. PROTOCOLS

(Ref.: P22-ET-09-000-001-0, Section 6 -Item “Protocolos a utilizar”)

To ensure seamless integration and reliable data exchange within the Ultra High-Pressure Reverse Osmosis (UHPRO) system, the following communication protocols have been considered:

- **EtherNet/IP** – Facilitates high-speed, deterministic communication between the PLC and field devices, supporting real-time control and diagnostics.
- **Modbus TCP/IP** – Enables robust and standardized communication between the PLC and the Client’s main control system, ensuring interoperability across platforms.
- **4-20mA with HART Protocol** – Provides analog signal transmission with digital overlay for enhanced device diagnostics and configuration, particularly for pressure and flow transmitters.

The final selection of protocols will be confirmed during the execution phase of the project. These decisions will be documented in the engineering submittals and will reflect the outcomes of technical discussions and mutual agreement on the finalized scope of work.

Technical Proposal

9. CONTROL PHILOSOPHY

(Ref.: P22-ET-09-000-001-0, Section 6 -Item “Filosofía de control”)

CONTROL PHILOSOPHY (Preliminary Description)

The Ultra High-Pressure Reverse Osmosis (UHPRO) train will have a PLC and operator interface that controls the on-skid valves and feed pump and monitors on-skid instrumentation. The UHPRO train will be designed to normally operate automatically.

OPERATOR INTERFACE AND SCREENS

The operator interface screens will be developed and provided by BW Water Americas (BWWA or BW Water) after system integration is carried out during the project execution phase.

Some actual screens will be then taken from the design software showing all the screen elements.

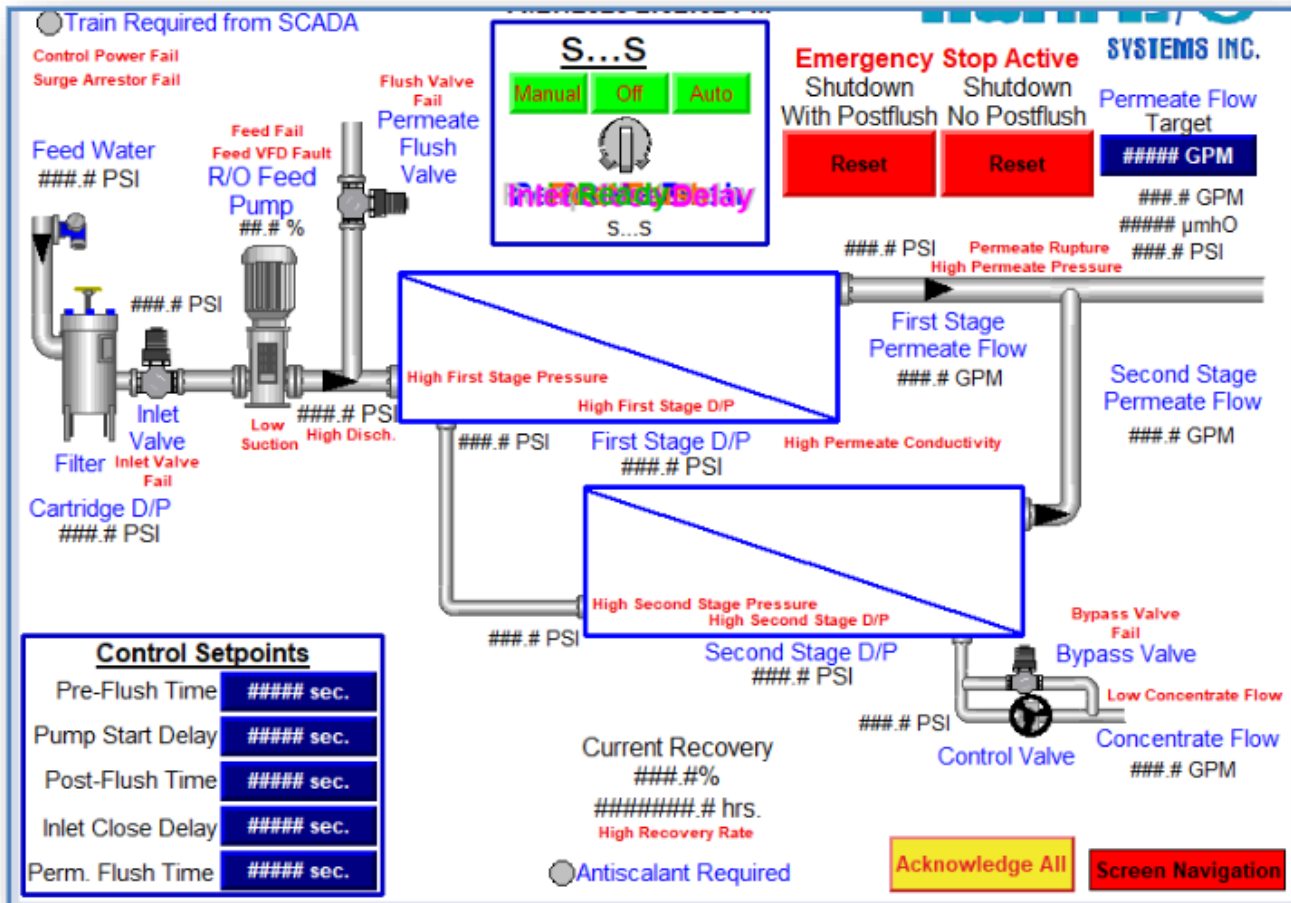
Overview (Primary) screen

- Displays the **operational status** of the UHPRO train.
- Hosts the **primary operator controls**.
- Displays **Valve status indicators**:
 - **Red icons**: Valve closed
 - **Green icons**: Valve open

A symbolic example from a non-UHPRO project is shown for reference; actual screens may differ after system integration.

Technical Proposal

Picture: Overview (primary) Screen - for reference only



Screen Navigation

- Accessed via the Screen Navigation button on the Overview screen.
- Provides access to:
 - Equipment Controls
 - Alarm Setpoints
 - Alarm History
 - Tech Support
 - Alarm Configuration
 - System Configuration

Technical Proposal

- Permeate Flush sequence can be toggled (enabled or disabled) from this screen, but only after entering the System Configuration Access Code (provided by BW Water).
- The System Configuration Access Code allows to access the Alarm Config and System Config screens.

Picture: Screen Navigation Sample - for reference only

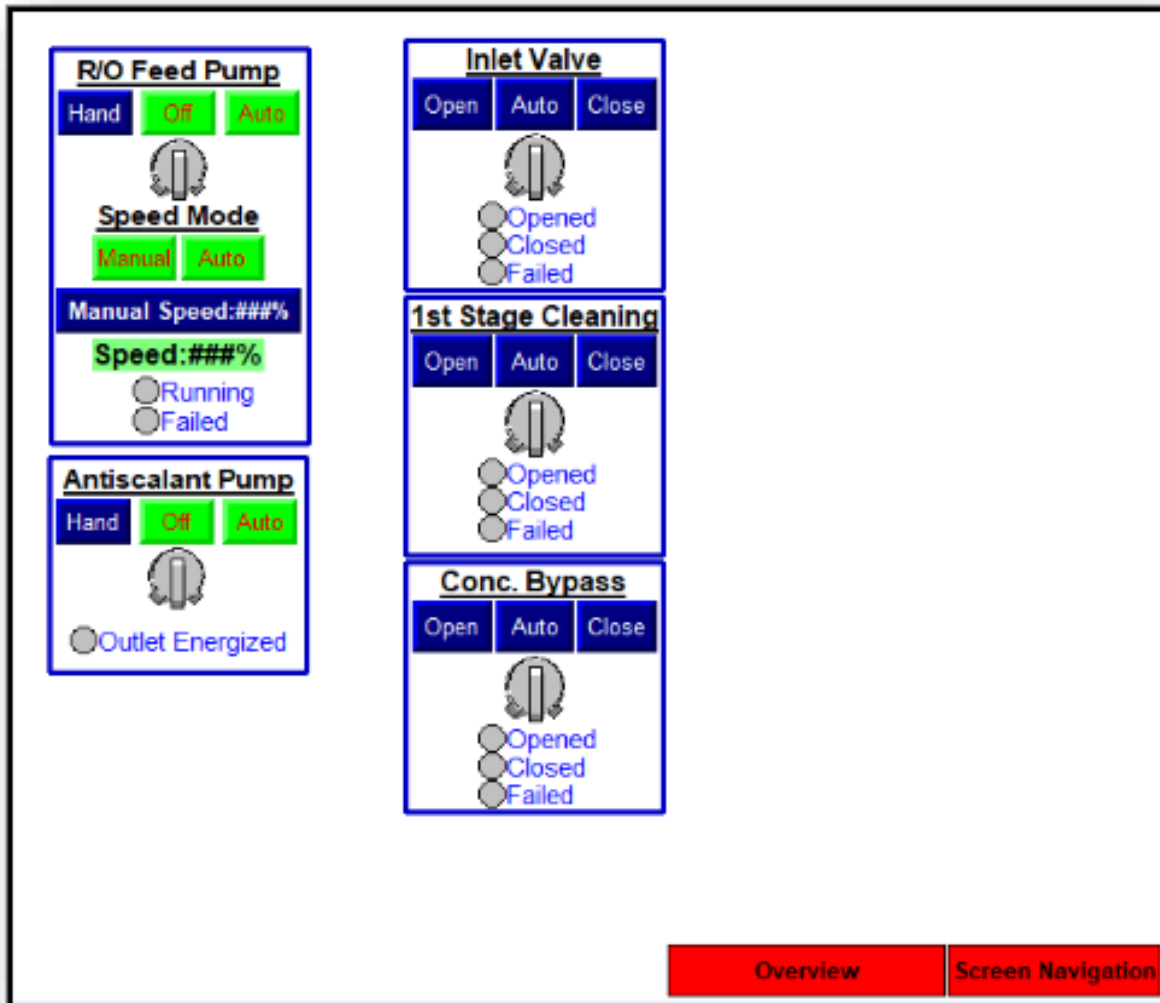


Equipment Controls

- Accessed via the Equipment Controls button on the Screen Navigation (first picture below) or by touching component icons on the Overview screen of these component control pop-up window (second picture below).
- Allows direct control of individual components.
- Used for maintenance and troubleshooting.

Technical Proposal

Picture: Equipment Controls Button Screen Sample - for reference only

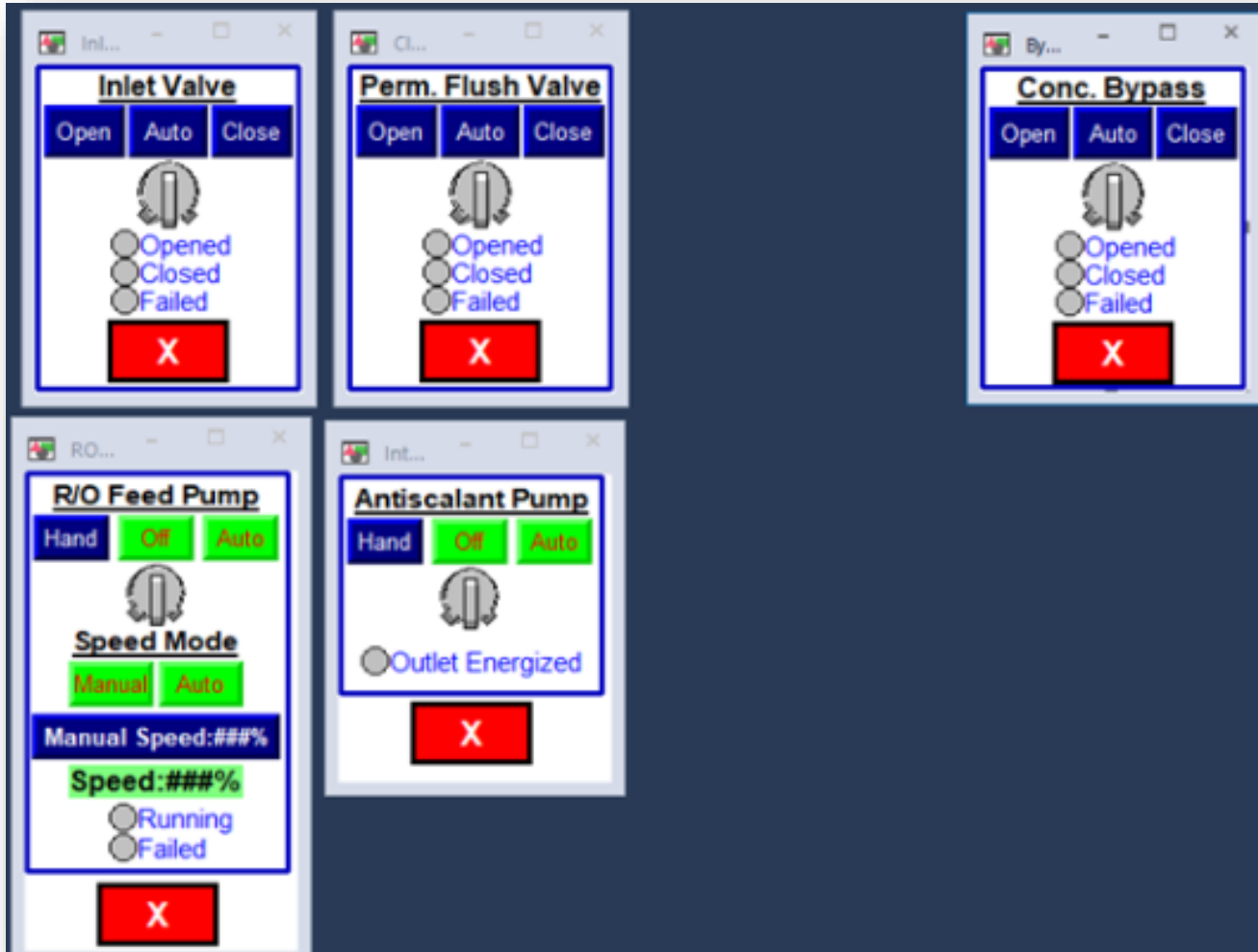


Component Control Pop-Up Windows

- Accessed via the Component icon on the Overview Screen (second picture below)
- During normal operation, all components should be set to Auto.

Technical Proposal

Picture: Component Control Pop-Up Windows Icon - for reference only



Alarm Setpoints

- Accessed via the Alarm Setpoints button on the Screen Navigation screen (see picture below)
- Used to configure alarm thresholds.

Technical Proposal

Picture: Alarm Setpoint Screen - for reference only

	<u>Alarm Setpoints</u>	<u>Current</u>
High Recovery %	###.##%	###.##%
Low Concentrate Flow	##### gpm	###.# GPM
High Total Perm. Conductivity	###.# μmho	##### μmho
High First Stage Press.	###.# PSI	###.# PSI
High Second Stage Press.	###.# PSI	###.# PSI
High First Stage D/P	###.# PSI	###.# PSI
High Second Stage D/P	###.# PSI	###.# PSI

Overview
Screen Navigation

Alarm History

- Accessed via the Alarm History button.
- Displays historical alarm data for diagnostics and review.

Technical Proposal

Picture: Alarm History Screen - for reference only

[illegible]

Tech Support

- Accessed via the Tech Support button.
- Requires a Hardware Configuration Access Code (provided by BW-Water).

Picture: Tech Support Screen Sample - for reference only

Hardware Configuration Access

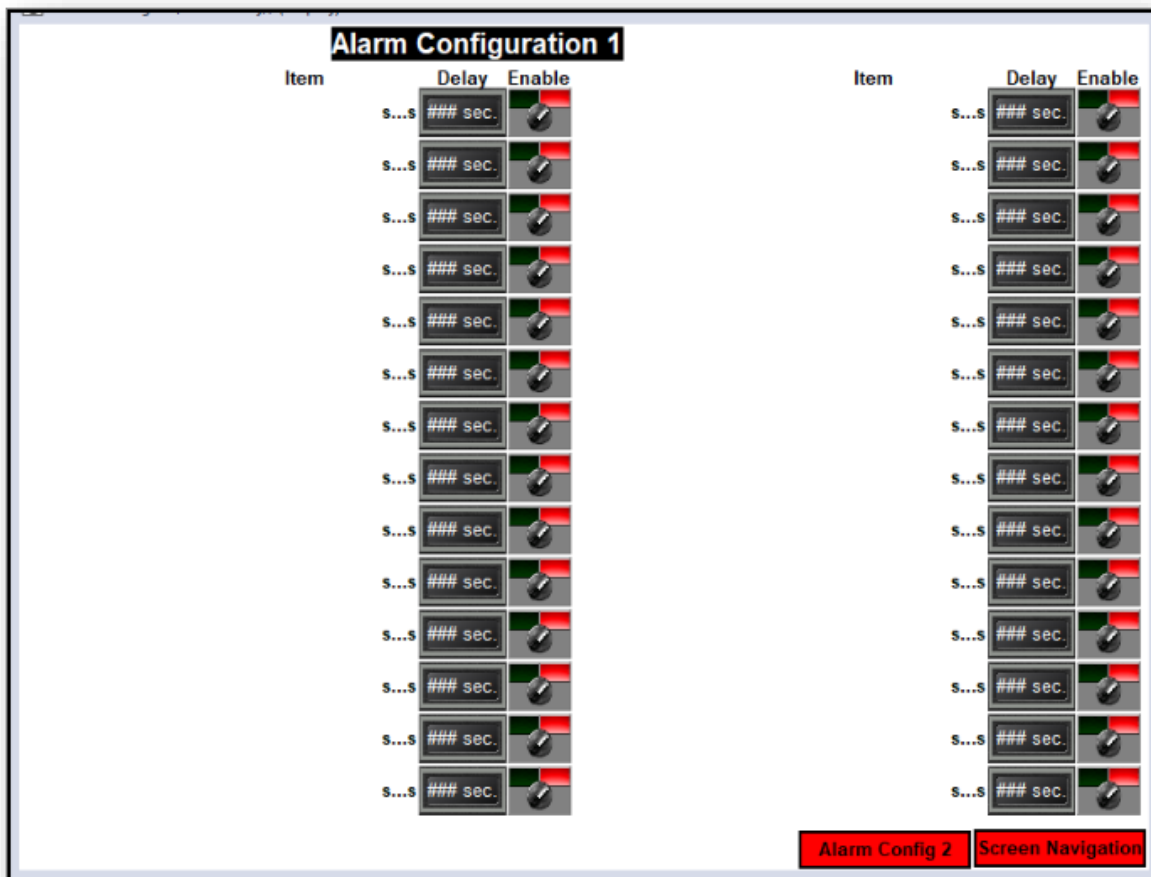
Screen Navigation

Technical Proposal

Alarm - Configuration 1

- Accessed via the Alarm Config button.
- Allows enabling/disabling alarms and adjusting delay times.
- Requires the System Configuration Access Code (provided by BW-Water).

Picture: Alarm Config.1 Screen - for reference only



Alarm - Configuration 2

- Accessed from the Alarm Config 1 screen by clicking Alarm Config 2 button
- Displays spare alarms.

Technical Proposal

Picture: Alarm Config.2 Screen - for reference only



Alarm Configuration 2

Item	Delay	Enable	Item	Delay	Enable
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐
\$...\$	### sec.	☐	\$...\$	### sec.	☐

Alarm Config 1 Screen Navigation

System Configuration

- Accessed via the System Config button on the Screen Navigation
- Allows adjustment of:
 - PID loop parameters
 - Analog instrument ranges
 - Operator interface date and time
- Requires the System Configuration Access Code (provided by BW-Water)

Technical Proposal

Picture: *System Configuration Screen - for reference only*

Feed Speed		S...S		S...S		S...S		S...S		S...S	
Proportional	##.##	#### mA	###.##	#### mA	###.##	#### mA	###.##	#### mA	###.##	#### mA	###.##
		Max Raw	Max Scl	Max Raw	Max Scl	Max Raw	Max Scl	Max Raw	Max Scl	Max Raw	Max Scl
		####	####.##	####	####.##	####	####.##	####	####.##	####	####.##
Integral	##.##	Min Raw	Min Scl	Min Raw	Min Scl	Min Raw	Min Scl	Min Raw	Min Scl	Min Raw	Min Scl
		####	####.##	####	####.##	####	####.##	####	####.##	####	####.##
Derivative	##.##										
Initial Speed	###.##										
Initial time	###.##										

S...S	
#### mA	###.##
Max Raw	Max Scl
####	####.##
Min Raw	Min Scl
####	####.##

#####.##

Reset Runtime

Permeate Flush

Enable ☒ Disable

Perm. Flush Fail Time

###

Screen Navigation

Date - Time Entry

Current Date and Time

Monday, November 27, 2023 2:03:35 PM

Change Time and Date to

Month	Day	Year
##	##	####
Hours	Minutes	Seconds
##	##	##

All fields will be updated

Set New Date and Time

Pre-Start-Up Review - – UHPRO Train

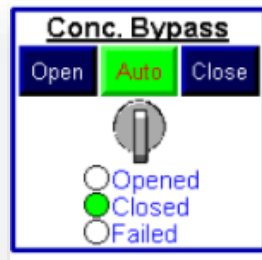
If the UHPRO train remains offline for an extended period due to chemical cleaning, maintenance, or repairs, the operator must perform a pre-start-up review before initiating UHPRO Train/system start-up.

The operator should conduct a thorough walk-through of the plant to verify that all equipment is ready for operation and that all manual valves and switches are correctly positioned.

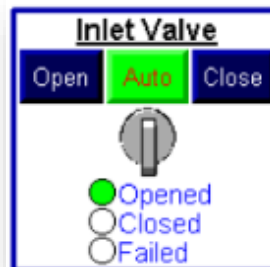
Pre-Start-Up Procedure:

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- Cartridge Filters should contain clean filters, and the manual inlet isolation valve should be open. The vent valves should be **open** especially if the cartridge filters have been changed to bleed any entrapped air. The clean water and dirty water drain valves should be **closed**
- Concentrate discharge valve must be in **automatic** mode.



- All manual valves off skid and downstream that could restrict the concentrate or permeate flow leaving the skid must be **opened**.
- UHPRO train inlet valve must be in **automatic** mode.



- The high-pressure feedwater pump must be in **automatic** mode.
- **Under low pressure flow from feedwater source, bleed air off from feedwater system, the cartridge filters, the UHPRO feed pump, Super Duplex stainless steel manifolds and permeate sample valves. If the skid is drained, the pre-flush timer needs to be extended, ensuring that all entrapped air is removed from the system.**

Pre-Flush, Normal Operation, Post-Flush, and Permeate Flush

- When the UHPRO train is called to run from SCADA, the train inlet valve will open.

Technical Proposal

- Once the train inlet valve is fully open the pre-flush timer will start, and water will flush through the system for the length of time input in the pre-flush timer setting.
- The pre-flush time can be monitored on the UHPRO train operator interface screen and measured in seconds remaining.
- When the Pre-Flush sequence will time out the UHPRO high pressure feed pump will slowly ramp up.
- The manual concentrate control valve will need to be adjusted if the correct concentrate flow is not displayed on the UHPRO train operator interface screen.
- After start-up, the UHPRO train should reach stable operation in approximately 5-10 minutes.
 - o **Feed Pump Speed** – In Automatic mode, this speed should adjust automatically to meet the Total Permeate flow set-point. Changing feed pump speed will change flows throughout the system including total permeate and concentrate flow. Increasing the feed pump speed will increase flows while decreasing it will lower the flows.
- The UHPRO system will continue operating until the UHPRO skid is commanded to stop by SCADA. At this time, the UHPRO train will go into a post-flush sequence.
- During post-flush, the high-pressure feed pump will be de-energized. The concentrate discharge valve will fully open, and concentrated water will be flushed from the system under pressure. The system will flush for the length of time input in the post-flush timer setting. The post-flush time remaining can be monitored and changed on the UHPRO train operator interface screen. At the end of the post-flush sequence the train inlet valve will close.
- The UHPRO train will remain shut down until it is called for by SCADA.

Post-flush will be required to reduce the conductivity of the water that will remain in contact with the membranes until the system is restarted. If the concentrate is not flushed out of the system, salts will begin to precipitate out of this saturated solution and scales will form on the membrane surface. The goal of the post-flush is to remove the water left in the pressure vessels similar in quality to that of the feed water. Feed water is naturally balanced meaning no salts present exceed the saturation limit and therefore salts will not precipitate out of solution. Opening the concentrate valve will allow for a high volume of water to be flushed through the system at low pressure. This will reduce the required post-flush length.

Permeate post-flush can be utilized to further reduce the conductivity of the water that will remain in contact with the membranes until the system is restarted. As soon as the UHPRO train is brought into operation and is producing permeate, it will fill the dual-purpose CIP tank with permeate. The tank will

Technical Proposal

fill using a float valve that will prevent further filling once the tank is at maximum capacity. Once the UHPRO train will fully complete the typical raw water post-flush the 1st stage cleaning feed valve will open. Once it is fully open, the UHPRO train will call for the CIP pump to run. The pump will run for a pre-determined amount of time before shutting down. The 1st stage cleaning feed valve will close.

Operation without SCADA – An operator can operate the UHPRO train without SCADA in an emergency situation under certain conditions. Sufficient feedwater must be available at the train inlet valve. When the UHPRO train control switch is placed into Manual the automatic operation sequence will commence without requiring communication with SCADA.

When the UHPRO train control switch is placed into Off the UHPRO train valves and feed pump can also be operated manually. This will require the operator to operate the components in the correct order and should only be done in an emergency situation as there is greater risk of unintended operation.

Note: When operating the UHPRO train manually, the train continues to run regardless of conditions, with most of the alarms non-functional. Each individual system component and valve in manual can only be stopped/started or opened/closed by the operator. This mode of operation disables all automatic control functions and interlocks. Only skilled technicians and/or operators with extensive knowledge of the process should operate the system manually.

Extreme caution must be utilized when operating the system manually in order to avoid potentially catastrophic damage to pumps, piping, controls, membranes and all other system components.

UHPRO Feed Pump Control – As stated above, the UHPRO feed pump speed will typically be controlled automatically when the Speed Mode is set to Auto in order to maintain the total permeate flow rate setpoint. The Speed Mode can be set to Manual to run at a set speed (Hz) for troubleshooting or maintenance purposes.

This functionality is useful for testing purposes when wanting to compare operational data versus start-up data to determine if a cleaning is necessary. The flow from each vessel can be measured with the 3-way permeate ball valves and volumetric measurement (calibrated bucket & stopwatch). Also, it will give the operator the ability to set the pump to a historical speed and run the system if a flow meter were to malfunction and need to be taken out of service.

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UHPRO Skid: Alarm Conditions – There will be several alarm conditions monitored by the UHPRO train PLC that will initiate either an immediate UHPRO train shutdown or a shutdown with post-flush and permeate flush if enabled. If the shutdown is from an alarm that results in an immediate shutdown, then the post-flush step will be skipped.

UHPRO Train Initiated Some Possible Alarms: Shutdown with Post-Flush

- High Recovery % Alarm (Operator Adjustable Setpoint)
- RO Feed Pump VFD Fault
- RO Feed Pump VFD Failed to Run

UHPRO Train Initiated Some Possible Alarms: Immediate Shutdown without Post-Flush

- Emergency Stop Switch
- Low Feed Pump Suction Pressure (Mechanically Adjusted)
- High Feed Pump Discharge Pressure (Mechanically Adjusted)
- Rupture Disk Burst (Requires Rupture Disk Replacement)
- Low Concentrate Flow (Operator Adjustable Setpoint)
- High First Stage Feed Pressure (Operator Adjustable Setpoint)
- High Second Stage Feed Pressure (Operator Adjustable Setpoint)
- High 1st Stage Differential Pressure (Operator Adjustable Setpoint)
- High 2nd Stage Differential Pressure (Operator Adjustable Setpoint)
- Inlet Valve Failed to Open
- Inlet Valve Failed to Close
- Concentrate Discharge Valve Fail to Open
- Concentrate Discharge Valve Fail to Close
- 1st Stage Cleaning Feed Valve Fail to Open
- 1st Stage Cleaning Feed Valve Fail to Close
- Surge Arrestor Fail
- Control Power Fail

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In all cases, if the UHPRO train shuts down in an alarm condition, the alarm text will flash on the operator interface next to the related component icon.

The UHPRO train will remain shut down until the operator has addressed the cause of the alarm and presses the alarm Reset button that appears on the operator interface.

When the alarm condition has been addressed and the alarm Reset button pushed, the UHPRO skid will begin a new running sequence starting with pre-flush if the UHPRO train control is still in Auto and SCADA is still calling for the UHPRO train to run. If it is not desired, then the UHPRO train control can be switched to Off.

If the alarm condition has not been cleared the alarm will not be able to be reset.

Alarms can be viewed on the Alarm History screen.

Technical Proposal

10. HARDWARE AND SOFTWARE LICENSES LIST

(Ref.: P22-ET-09-000-001-0, Section 6 -Item "Listado con licencias de hardware y software")

Hardware / Software Licenses (Preliminary List)						
Item	Description / Model Number	Type	Manufacturer	Location	Qty	Notes
1	Studio 5000 Logix Designer v37	Logic Editor	Rockwell Automation	-	1	
2	FactoryTalk View Studio v15	HMI Editor	Rockwell Automation	-	1	
3	Dell 14" Latitude 3450 (w/ Win 11)	Laptop	Dell	-	1	

Technical Proposal

11. PROJECT SCHEDULE

(Ref.: P22-ET-09-000-001-0, Section 6 -Item “Programa del encargo”)

The final project execution schedule will be established based on a mutually agreed timeline following contract finalization. However, the following **preliminary milestones** are proposed based on current requirements and conditions:

- **Issuance of Award Notice:** September 29, 2025
- **Start Date of contract:** will be **15 - 20 days after Award Notice**. This is when the contract is expected to be fully executed, and the final agreed timeline shall be counted from this date
- **Contract Term:** Shall not exceed **510 calendar days from start date until Provisional Acceptance**
- **Delivery Period:** Maximum of **300 calendar days from start date**
- **Engineering submittals for approval:** overall maximum **90 calendar days** after the **start date**, except in cases where difference period is defined. Please refer to Section 11.1 below of this proposal.
- **Fabrication, assembly and testing activities:** start after selected/required documentation is approved
- **Shipment of system:** after ADASA issued the release, which required all applicable engineering submittals to be approved. And agreed list to be defined
- **Installation supervision Commissioning and Start-Up Period:** Maximum of **21 calendar days**, initiated upon ADASA’s formal request. This request (advance notice to site) can be made by ADASA within **180 calendar days** following receipt of the UHPRO system on site. Note:
- **Advance Notice for On-Site Field Services:** ADASA will notify the supplier **30 calendar days in advance** of the required site appearance date.
- **Final Documentation and Project Close-Out:** Supplier will have a maximum of **30 calendar days** after completion of commissioning and start-up activities to submit all required documentation and fulfill all conditions necessary for **Provisional Acceptance**
- **Review/approval Period:** All submittals shall be reviewed and approved by ADASA within 7 business days of submission
- **Warranty period:** 24 months from the Provisional Acceptance date (as per client requirement).
Note: for RO Membranes, the warranty offered will be as per manufactures and is not part of this Equipment guarantee)

Technical Proposal

- **Final Acceptance:** after warranty period has satisfactorily elapsed. The Supplier may request the ADASA to issue the Final Acceptance Certificate in writing. Upon formalization of the Certificate, the ADASA Buyer will return the corresponding Warranty Certificates to the Supplier.

A visual timeline (Gantt-style chart) for the Preliminary Project Execution Schedule is attached for further details.

11.1 ENGINEERING SUBMITTALS & SCHEDULE

(Ref.: P22-ET-09-000-001-0, Section 6 -Item "Listado de entregables de ingeniería")

A final list of engineering submittals and timeline will be established based on a mutually agreed agreement following contract finalization. According to P22-ET-09-000-001-0 -Section7, the following has been considered at this stage.

Engineering Submittals						
<u>References:</u> G = General P = Process M = Mechanical E = Electrical I&C = Instrumentation & Control Q = Quality (Assurance / Control)						
Doc. No.	Discipline	Document Name	Submittal Due			Notes
			#	Unit	Event	
TBA-1	G	Detailed Project Schedule, including engineering, procurement, construction, transportation, assembly, and commissioning.	15	Days	after the start date ,	after the start date , as defined above
TBA-2	P	Process and Instrument Diagrams (P&IDs)	within 90	Days	after the start date	
TBA-3	P	Process Flow Diagram (PFD) with Mass Balance	within 90	Days	after the start date	
TBA-4	P	Process calculations reports with models using the selected membranes for 53,000 mg/L TDS - o RO Projections - o Turbo Charger Projections - o Utility Consumption List - o Chemical Consumption List	within 90	Days	after the start date	
TBA-5		General arrangements (GAs)/Layouts Drawings	within 90	Days	after the start date	
TBA-6		Plan and elevation drawings of equipment and piping	within 90	Days	after the start date	
TBA-7		Module Seismic Calculation Report	within 90	Days	after the start date	

Technical Proposal

Engineering Submittals						
References: G = General P = Process M = Mechanical E = Electrical I&C = Instrumentation & Control Q = Quality (Assurance / Control)						
Doc. No.	Discipline	Document Name	Submittal Due			Notes
			#	Unit	Event	
TBA-8		Stress & flexibility analysis for high pressure pipes	within 90	Days	after the start date	
TBA-9		Drawing with civil requirements, including dimensions, loads, and bolts	within 90	Days	after the start date	Only GAs with Weight and Dimensions, Client shall obtain the require civil design
TBA-10		Single-line diagrams	within 90	Days	after the start date	
TBA-11		Main Equipment data sheets (defined in P22-ET-09-000-001, Section 5.1) - High Pressure Pump 1st Stage Turbo 2nd Stage Turbo - CIP Wash Pump - Cartridge Filters - RO Pressure Vessels - Reverse Osmosis Membranes - Dosing Pump - RO Support/skid Frame - Container - Auxiliary Services (container interior lighting, exterior lighting, A/C	within 90	Days	after the start date	
TBA-12		Manufacturing and testing dossier for all electrical and electromechanical equipment with their special steel manufacturing certificates for super duplex equipment.	within 90	Days	after the start date	
TBA-13		Piping, Valves and instruments data sheets/specs, with brands and models	within 90	Days	after the start date	
TBA-14		Control philosophy with reference to the P&ID TAGs	within 90	Days	after the start date	
TBA-15		Design of control screens and HMI.	within 90	Days	after the start date	